Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 6: Prokaryotes

- Prokaryotic cells are simpler than eukaryotic cells: no membrane-bound nucleus, no membrane-bound organelles, no microtubules and simpler flagella.

6.1 Taxonomy and phylogenetic position (SLO 1, 2)

- **R. H. Whittaker** (1969) proposed the five-kingdom system distinguishing prokaryotes from eukaryotes; **Margulis and Schwartz** (1988) revised it and designated kingdom **Monera** for all prokaryotes.

Phylogenetics: Study of evolutionary relatedness among groups of organisms. Phylogeny = origin and evolution of a set of organisms.

- Prokaryotes form two domains: **Bacteria** (Eubacteria) and **Archaea** (Archaebacteria); with **Eukarya** they make the **three-domain system**. **Carl Woese** (1977) showed archaea have a separate evolutionary lineage.
- Ancestors of bacteria appeared about **4 billion years ago**; for about 3 billion years all life was microscopic. Last common ancestor of bacteria and archaea lived about 2.5-3.2 billion years ago, probably a hyperthermophile.

6.2 Archaea

- Greek archaios = ancient. Live in extreme environments like early earth. Special membrane lipids for extreme heat / acid; **no peptidoglycan** in cell walls; some make **methane**; many make food from H_2, S, CO_2, water; different rRNA.

Feature	Eubacteria	Archaea
Peptidoglycan in cell wall	Present	Absent
RNA polymerase	Relatively small and simple	Complex, similar to eukaryotes
Introns	Absent	Present in some genes

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Feature	Eubacteria	Archaea
Membrane lipid carbon chains	Unbranched	Branched
rRNA / ribosomal proteins	Many unique	Many match eukaryotes

Group of archaea	Habitat and example
Methanogens	Methane producers; anaerobic swamps, marshes, intestines. Methanobacterium smithii (human gut).
Halophiles	Salt lovers; salted fish and meat. Halobacterium halobium.
Thermoacidophiles	Hot and acidic: hot springs, geysers, volcanic areas. Thermoplasma acidophilum, Thermus aquaticus.

6.3 Bacteria: ecology and diversity (SLO 3, 5)

- Study of bacteria = **bacteriology** (branch of microbiology). **Leeuwenhoek** (1674) first saw bacteria ("animalcules") with a single-lens microscope. **Ehrenberg** (1882) named them bacterium (Greek baktria = little stick).

Why bacteria occur in the widest range of habitats (SLO 3)

1. **Flagella:** motility to move toward favourable and away from unfavourable places.
2. **Pili:** hair-like structures for adhesion to each other and to surfaces (rocks in streams, human intestine) for colonization and nutrition.
3. **Endospores:** survive lack of water and nutrients, heat, cold and most poisons; germinate when conditions improve.
4. **Actinomycete filaments:** branching cell chains in soil; bridge dry gaps between soil particles and break down organic matter.
5. Diverse nutrition (autotrophic, heterotrophic) and respiration (aerobic, anaerobic).

Major group	Features / examples
Green non-sulphur	Mostly photosynthetic, some thermophilic; Chloroflexus.
Green sulphur	Phototrophic, no oxygen produced; Chlorobium.

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Major group	Features / examples
Proteobacteria (purple)	Gram negative; some phototrophic (no O_2), some chemotrophic, some nitrogen fixers; Escherichia, Salmonella.
Gram positive	Endospore formers, lactic acid bacteria, cocci; Staphylococcus, Clostridium, Bacillus, Mycoplasma.
Cyanobacteria	Phototrophs producing oxygen; Nostoc, Anabaena, Oscillatoria.
Bacteroides	Gram negative rods; Bacteroides, Flavobacterium.

Cyanobacteria - most prominent photosynthetic bacteria (SLO 5)

- Blue-green algae, Gram negative. Found in damp places, salt and fresh water, moist soil, rocks, tree trunks, hot springs up to 85 C; epiphytic or symbiotic (liverworts, ferns, corals; with fungi = lichens). Among the first colonizers of land.
- Forms: unicellular, colonial, filamentous. Prokaryotic features: no nuclear membrane, no protein with DNA, no membrane organelles, cell wall of **muramic acid** (only in prokaryotes) often with mucilage sheath, circular DNA, many ribosomes.
- **Photosynthesis** in extensive membrane system in outer cytoplasm; **oxygen released**; accessory pigment **phycobilins** (blue **phycocyanin** predominant). Believed to have first introduced oxygen into the primitive atmosphere.
- About one-third **fix atmospheric nitrogen** in **heterocysts** (thick walled, no nuclei). In Pakistan **Nostoc** and **Anabaena** are cultivated to increase soil fertility.
- Reproduction: no sexual; asexual by cell division, **fragmentation** (hormogonia, near heterocyst), **akinetes** (thick walled resting cells with food and DNA), rarely endospores from heterocyst.

6.4 Structure of bacteria (SLO 4, 6, 7, 8)

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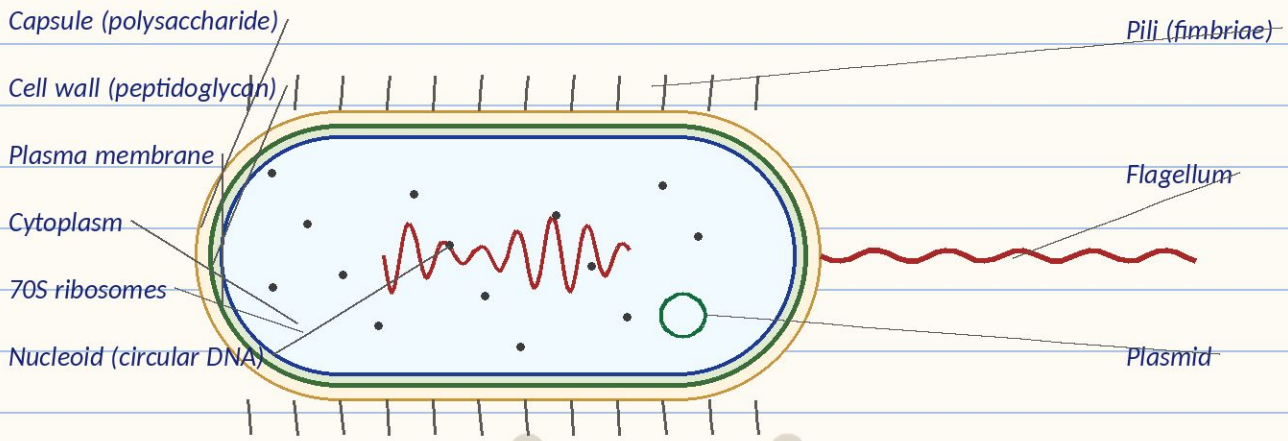


Fig: Generalized bacterial cell

- *Typical bacterium: cell wall, cell membrane, cytoplasm, genetic material and surface structures (capsule, flagella, pili). Cell wall is absent in **Mycoplasma** (only membrane).*
- ***Cell wall:** outermost layer common to all bacteria; inner **peptidoglycan** (murein / mucopeptide, found only in bacterial walls; peptide + glycan) gives shape and support; outer membrane varies.*
- ***Capsule:** gelatinous layer of polysaccharide (anthrax bacillus: polymerized D-glutamic acid); helps adherence to host tissues.*

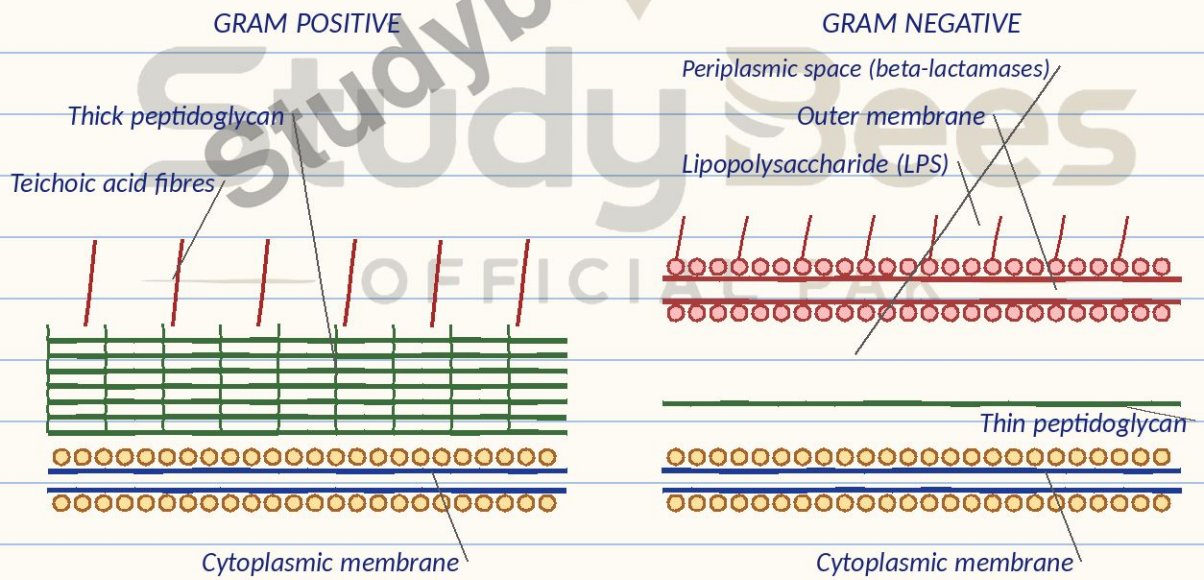


Fig: Cell wall of Gram positive and Gram negative bacteria

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Gram positive	Gram negative
Thick peptidoglycan layer	Thin peptidoglycan layer
Some have teichoic acid fibres projecting outside	No teichoic acid
No outer membrane	Outer membrane of lipopolysaccharides, lipoproteins, phospholipids
No periplasmic space	Periplasmic space between outer and cytoplasmic membrane (beta-lactamases break penicillin: antibiotic resistance)
Retain violet (crystal violet) in Gram stain	Lose violet; stain pink / red

Diversity of shapes and sizes (SLO 8)

- Three main shapes: **spherical** (cocci), **rod shaped** (bacilli), **spiral** (spirochetes, spirilla). Size 0.1 to 600 micrometre in one dimension.

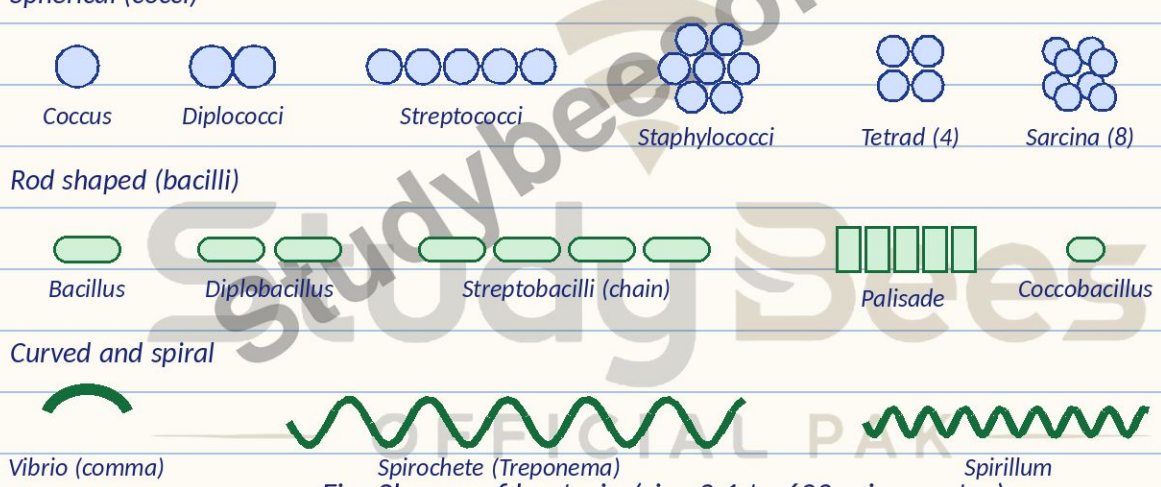


Fig: Shapes of bacteria (size 0.1 to 600 micrometre)

- **Cocci:** diplococci (pairs), streptococci (chains), staphylococci (grape-like clumps), tetrad (4), sarcina / octad (8). Examples: Streptococcus pneumoniae, Neisseria meningitidis.
- **Bacilli:** diplobacillus (pairs), coccobacilli (short ovoid), vibrio (comma), palisade (coin stack). Examples: Escherichia coli, Pseudomonas.
- **Spirochetes:** thin walled flexible spirals, usually single; Treponema pallidum.

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Endospore formation (SLO 9)

- ***Sporulation:*** *one bacterium forms one spore containing DNA, a little cytoplasm, ribosomes, peptidoglycan, very little water and a thick keratin-like coat. Seen in* ***Clostridium*** *and* ***Bacillus****. Survives high temperature, desiccation and chemicals; germinates into a vegetative cell when conditions improve.* ***Not reproduction*** *(no increase in cell number).*

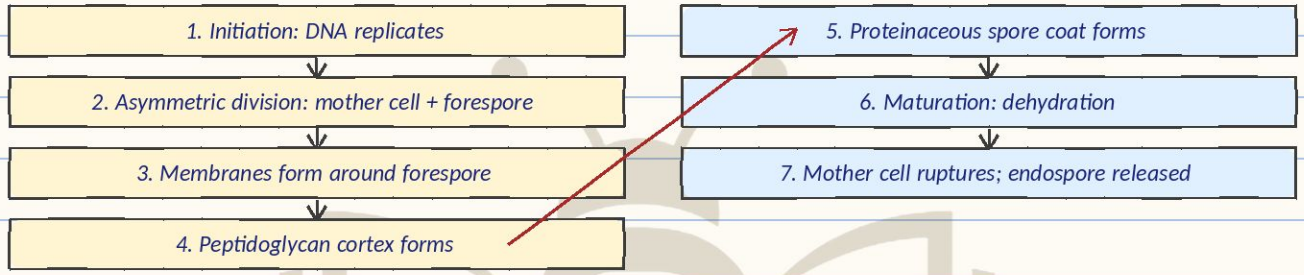


Fig: Endospore formation (not reproduction)

Motility and flagella (SLO 10, 11)

- *Bacteria move by* ***flagella****, gliding,* ***twitching*** *(pili used like grappling hooks) or buoyancy changes; spirochetes twist their helical bodies.*
- *Flagella arrangement:* ***monotrichous*** *(one),* ***amphitrichous*** *(one at each end),* ***lophotrichous*** *(tuft at pole),* ***peritrichous*** *(over whole surface).* ***E. coli*** *moves by swimming and tumbling (3D random walk). Movement toward or away from stimuli =* ***taxes****: chemotaxis, phototaxis, magnetotaxis.*

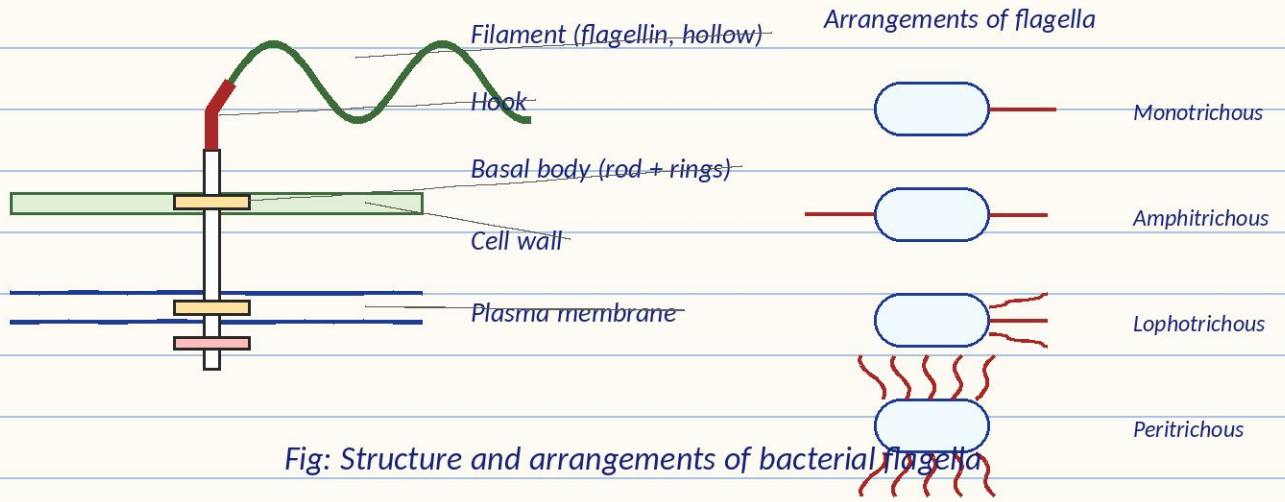


Fig: Structure and arrangements of bacterial flagella

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- ***Flagellum:*** *thread-like outgrowth, one protein* ***flagellin*** *(non-contractile, no microtubules), about 20 nm wide and up to 20 micrometre long. Three parts:* ***basal body*** *(just under the membrane; makes rotatory motion),* ***hook*** *(joins basal body to filament) and* ***filament*** *(hollow, helical protein chains). 360 degree rotation spins the cell forward.*

Genomic organization (SLO 12)

- ***E. coli:*** *one circular DNA molecule of about 5 million base pairs coding for about 2000 proteins. Most bacteria have a single circular chromosome (160,000 base pairs in Carsonella to 12.2 million in Sorangium). Exception:* ***Borrelia*** *spirochetes have a linear chromosome.*
- *Many bacteria also carry* ***plasmids*** *(accessory rings of DNA).*

6.5 Modes of nutrition (SLO 13, 14)

- *By energy and carbon source:* ***autotrophic*** *(fix CO_2) or* ***heterotrophic*** *(organic carbon).*

Type	*Details and examples*
Photosynthetic autotrophs	*Bacteriochlorophyll in mesosome membranes / cytoplasm; use light, CO_2 and H_2S (not water). $2H_2S + CO_2$ to $(CH_2O) + H_2O + 2S$. Green sulphur, purple sulphur, purple non-sulphur bacteria.*
Chemoautotrophs	*Colourless; oxidize inorganic substances (sulphur, ammonia); energy used to make carbohydrate. Nitrifying and sulphur bacteria.*
Saprotrophic heterotrophs	*Feed on dead organic matter with extracellular enzymes; release nutrients. Pseudomonas, Azotobacter.*
Parasitic heterotrophs	*Obligate (Rickettsia) or facultative (E. coli); toxins and virulence factors. Streptococcus pneumoniae, Salmonella typhi, Mycobacterium tuberculosis, Vibrio cholerae.*

- ***Respiration:*** *aerobic (Pseudomonas), anaerobic (Spirochaeta), facultative (E. coli: with or without oxygen), microaerophilic (low oxygen, Campylobacter).*

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Cyanobacteria	Other photosynthetic bacteria (purple, green)
Complex apparatus; chlorophyll a, (b), phycobiliproteins (phycocyanin, phycoerythrin); wide light range	Bacteriochlorophylls (chlorosomes with bacteriochlorophyll c in green sulphur); absorb infrared / near infrared
In thylakoids (stacked like grana)	In cell membrane infoldings: chromatophores; no thylakoids
Water split; oxygen released	No oxygen; use sulphur compounds as electron donor

6.6 Growth and reproduction (SLO 15, 16)

- Binary fission gives **exponential (logarithmic) growth**: cells 1, 2, 4, 8, 16 = 2^n; one bacterium gives 16 after four generations. **Generation (doubling) time:** 20 minutes for E. coli to over 24 hours for M. tuberculosis (depends on species, nutrients, temperature, pH).

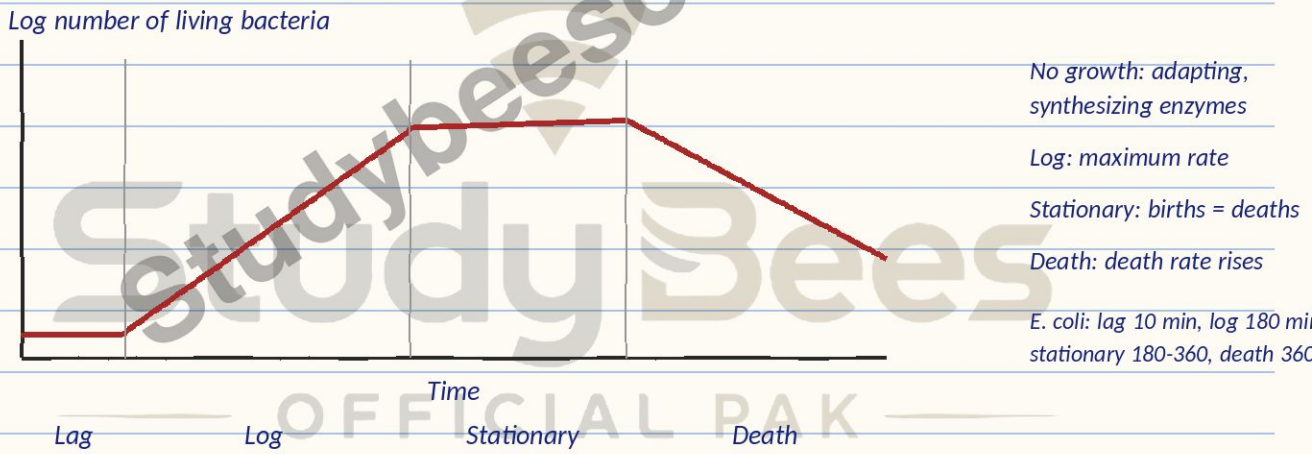


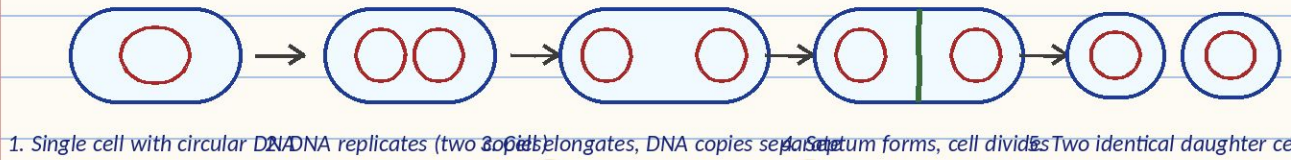
Fig: Growth curve of a bacterial population

Phase of growth	Description
Lag	Adapting to new environment; making new enzymes; growth not yet at maximum rate (E. coli 10 min).
Log (exponential)	Maximum growth rate, straight line on log graph (180 min).
Stationary	Growth rate zero: births equal deaths; nutrients exhausted, toxic wastes, oxygen depleted (180-360 min).

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Phase of growth	Description
Death (decline)	Death rate rises and cells stop multiplying (360-1800 min).

Binary fission (asexual)



Doubling time: 20 minutes (E. coli) to over 24 hours (M. tuberculosis)

Fig: Binary fission in bacteria

- DNA replication from the origin, bidirectionally; cell elongation (new wall and membrane); cell division (membrane pinches in, **septum** forms); two genetically identical daughter cells. E.g. E. coli, Staphylococcus aureus.

Sexual reproduction = genetic recombination

- DNA of two bacteria combines to give a new **recombinant**; by conjugation, transduction and transformation.

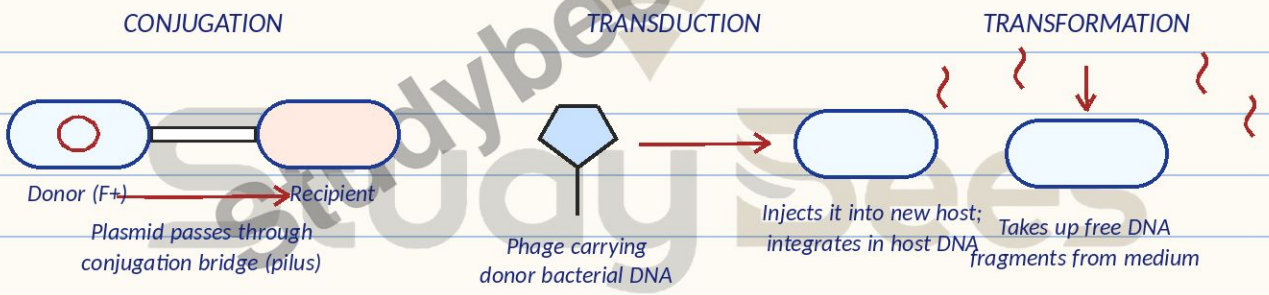


Fig: Genetic recombination in bacteria

- **Conjugation:** transfer of a plasmid (sometimes some chromosomal genes) from **donor** to **recipient** through a bridge (pilus). **Lederberg and Tatum** (1946, Nobel Prize): two X-ray induced mutants of E. coli (one unable to make substance A, the other needing substance B) mixed on medium with A and B gave offspring that grew on minimal medium (wild type) = genetic recombination; later seen directly with the electron microscope.

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- ***Transduction:*** *a bacteriophage carries bacterial DNA from one bacterium to another; bacterial genes integrate in the new host DNA and replicate with it.*
- ***Transformation:*** *a cell absorbs free DNA fragments released by dead or rapidly dividing donor cells; the recipient becomes a transformed cell.*
- *Bacteria are clonal but evolve through recombination and* ***mutations*** *(replication errors, mutagens, or stress-directed mutation).*

6.7 Importance of bacteria (SLO 17, 18)

Nutrient recycling

- ***Rhizobium*** *in legume root nodules fixes N_2 to ammonium; saprotrophs break down nitrogenous waste to ammonia;* ***Nitrosomonas*** *and* ***Nitrobacter*** *convert ammonia to nitrate (****nitrification****); denitrifying bacteria return nitrogen gas (****denitrification****). Fermentation releases CO_2 that plants use.*

Ecological importance

- *Bacteria and fungi are* ***decomposers****: dead plants and animals become* ***humus****, which enriches soil and holds moisture.*

Economic importance (beneficial)

Use	*Details*
Digestion	*Gut bacteria emulsify fats; cattle bacteria digest cellulose and starch.*
Vitamins	*Vitamin K and B12 (E. coli).*
Industry	*Acetic acid (vinegar), acetone, lactic acid, butanol, vitamins, cheese, tobacco flavouring; coffee and leather tanning.*
Food	*Yogurt, cheese, butter.*
Nutrient source	*Bacteria digested in animal gut give amino acids and vitamins.*
Antibiotics	*Streptomycin, aureomycin, terramycin (mostly from actinomycetes).*
Biogas	*Methane (54-70%) from sewage, dung, garbage.*
Single cell protein	*Protein from large-scale culture of bacteria, yeast, fungi, algae for food or feed.*

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Use in research and technology (SLO 18)

- ***Research:*** *bacteria, yeast, Neurospora widely used;* ***Hershey and Chase*** *(1952) used T2 bacteriophage and bacteria to prove DNA is the hereditary material.*
- ***Technology:*** *transgenic bacteria to clone genes or mass produce insulin, growth hormones, clotting factor VIII; frost-minus bacteria protect plants;* ***bioremediation*** *(oil-eating bacteria clean oil spills); chemical production (phenylalanine); mineral processing (copper, uranium, gold from low-grade ores).*

Harmful bacteria

- *Turn wine to vinegar; decay wood, leather, fabrics; spoil food; cause plant and human diseases.*

Bacterial diseases in man (SLO 19)

Disease	Causative agent, symptoms, treatment, prevention
Cholera	Vibrio cholerae. Profuse watery diarrhoea. Replace fluids and electrolytes (oral / IV); tetracycline shortens illness. Prevention: clean water supply, public health, killed or live vaccine (limited use), detect carriers.
Typhoid	Salmonella typhi. Slow onset, fever, constipation, high fever, delirium, tender abdomen, enlarged spleen, rarely rose spots. Antibiotics for chronic carriers. Prevention: hand washing, pasteurization of milk, proper cooking, vaccine.
Tuberculosis	Mycobacterium tuberculosis. Pulmonary TB: mild fever 7-14 days, dry cough; secondary: evening fever, night sweats, weight loss, blood-stained sputum, chest ache. Multiple-drug therapy (6-9 months) or DOTS. Prevention: prompt treatment of patients, masks, BCG vaccine (M. bovis).
Pneumonia	Streptococcus pneumoniae. Sudden chills, cough, pleuritic pain, rusty sputum. Penicillin and erythromycin.

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Bacterial diseases in plants (SLO 20)

Type	Symptoms, causes, prevention
Leaf spot	Discrete or spreading lesions on leaf blade (toxins). Xanthomonas campestris (tomato, pepper), Pseudomonas syringae (tobacco), ring disease of potato. Prevent contact; disease-free seeds.
Blight	Rapid necrosis, often from leaf tip downward; plant killed. Xanthomonas oryzae (rice), Erwinia amylovora (fire blight of pear, apple). Disease-free seeds, remove diseased plants.
Soft rot	Pectolytic / cellulolytic enzymes macerate tissue; wet, soft, fast spreading. Erwinia atroseptica (potato). Remove diseased plants.
Wilt	Blocked water transport; leaves lose turgidity, limp, dry. Pseudomonas solanacearum (potato), Xanthomonas. Disease-free seeds, right planting date, spacing.
Galls	Localized outgrowths. Rhizobium (root nodules), Pseudomonas savastanoi (olive knot), Agrobacterium tumefaciens (large galls), Xanthomonas (cotton). Crop rotation, remove diseased plants.

6.8 Normal flora (SLO 21, 22)

Normal flora: Bacteria (and fungi) that are permanent residents of certain body parts, especially skin, oropharynx, colon and vagina. Bacterial cells outnumber human cells about ten times.

Member	Location
Clostridium species	Colon
Escherichia coli and other coliforms	Colon, vagina
Lactobacillus species	Mouth, colon, vagina
Enterococcus faecalis	Colon

- **Benefits:** digestion and absorption (vitamins B and K, biotin, folate); immune support and regulation; protective barrier (competitive exclusion); short-chain fatty acids (butyrate); gut-brain axis (mood); vaginal health (lactobacilli keep it acidic); detoxification; metabolism and weight regulation; immune

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development in early life; restoring gut balance after antibiotics; homeostasis.

6.9 Control of harmful bacteria (SLO 23)

Chemical method	Details
Disinfectants	On surfaces / objects: bleach (sodium hypochlorite), hydrogen peroxide, quaternary ammonium compounds.
Antiseptics	On living tissue: alcohol hand sanitizers, iodine, hydrogen peroxide.
Preservatives	In food and cosmetics: sodium benzoate, potassium sorbate, parabens.
Water treatment	Chlorine and chloramine.
Chemotherapy	Sulfonamides, nitrofurans when antibiotics fail.
Physical method	**Details**
Sterilization	Steam, dry heat, gas, filtration, radiation; kills all life; surgical instruments, milk, meat.
High temperature	Moist heat coagulates protein; dry heat oxidizes constituents.
Radiation	Below 300 nm; gamma rays sterilize.
Membrane filter	For heat-sensitive antibiotics and serums.
Pasteurization	Pasteur: 71 C for 15 seconds or 62 C for 30 minutes; kills non-spore formers; keeps taste; reduces typhoid and TB risk.
Low temperature / freezing	Storage at 10-15 C; freezing at -10 to -18 C preserves meat, vegetables.
Drying	Dehydration prevents growth: dried milk, dried meat.

STEAM activity 6.1: antibiotic sensitivity

- Pour nutrient agar in petri dishes, divide plates, inoculate sections with swabbed samples (hands, surfaces), apply different antibiotics, incubate for a few days, record colony size and effectiveness; shows the importance of responsible antibiotic use and antibiotic resistance.

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Chapter 6: Short Questions with Answers (SLO based)

Q1. [SLO 1] Who introduced the five-kingdom system? Which kingdom holds prokaryotes?

Ans: R. H. Whittaker (1969). Margulis and Schwartz (1988) designated kingdom Monera for all prokaryotes.

Q2. [SLO 1] Name the domains of the three-domain system.

Ans: Bacteria (Eubacteria), Archaea and Eukarya.

Q3. [SLO 2] Define phylogenetics and phylogeny.

Ans: Phylogenetics: study of evolutionary relatedness of groups of organisms. Phylogeny: origin and evolution of a set of organisms.

Q4. [SLO 2] What was Carl Woese's contribution?

Ans: In 1977 he showed that archaea have a separate evolutionary lineage from bacteria.

Q5. [SLO 2] When did bacteria originate? What was the common ancestor like?

Ans: About 4 billion years ago; the last common ancestor of bacteria and archaea (2.5-3.2 billion years ago) was probably a hyperthermophile.

Q6. [SLO 1] Why are archaea so named? Write their unifying features.

Ans: Archaios = ancient. Special membrane lipids, no peptidoglycan, some make methane, many make food from H_2, S, CO_2, water; different rRNA.

Q7. [SLO 1] Write three differences between Eubacteria and Archaea.

Ans: Eubacteria have peptidoglycan, simple RNA polymerase and unbranched membrane lipid chains; archaea lack peptidoglycan, have complex RNA polymerase like eukaryotes, introns in some genes and branched lipid chains.

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Q8. [SLO 1] Name the groups of archaea with examples.

Ans: Methanogens (Methanobacterium smithii), halophiles (Halobacterium halobium), thermoacidophiles (Thermoplasma acidophilum, Thermus aquaticus).

Q9. [SLO 1] Who first discovered bacteria? Who named them?

Ans: Leeuwenhoek in 1674 ("animalcules"); Ehrenberg named them bacterium in 1882 (Greek: little stick).

Q10. [SLO 3] How do flagella and pili help bacteria to survive in wide habitats?

Ans: Flagella give motility toward favourable conditions; pili help adhere to surfaces (rocks, intestine) for colonization and nutrition.

Q11. [SLO 3] What are actinomycetes and where are they found?

Ans: Bacteria with branching cell chains / filaments, common in soil; they break down organic matter and bridge dry gaps between soil particles.

Q12. [SLO 5] Why are cyanobacteria called blue-green algae?

Ans: They are blue-green in colour because of phycocyanin (blue) with chlorophyll, and look like algae, but are prokaryotes (Gk kyanos = blue).

Q13. [SLO 5] Why are cyanobacteria the most prominent photosynthetic bacteria?

Ans: They have chlorophyll a, phycobilins, thylakoids, split water and release oxygen like plants; they introduced oxygen into the primitive atmosphere.

Q14. [SLO 5] Write the prokaryotic features of cyanobacteria.

Ans: No nuclear membrane, DNA not combined with protein, no membrane bound organelles, muramic acid in cell wall, circular DNA, many ribosomes.

Q15. [SLO 5] What are heterocysts and akinetes?

Ans: Heterocysts: thick walled cells without nuclei that fix nitrogen. Akinetes: enlarged thick walled resting cells with food and DNA.

Q16. [SLO 5] Why are Nostoc and Anabaena cultivated in Pakistan?

Ans: They fix atmospheric nitrogen and so increase soil fertility.

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Q17. [SLO 5] How do cyanobacteria reproduce?

Ans: Only asexually: cell division, fragmentation (hormogonia), akinetes and rarely endospores of heterocysts.

Q18. [SLO 6] What is peptidoglycan? What are its synonyms?

Ans: Interwoven macromolecule of peptide and sugar (glycan) found only in bacterial cell walls; also called murein or mucopeptide.

Q19. [SLO 6] What is the capsule? What is it made of?

Ans: A gelatinous layer around the bacterium made of polysaccharide (polymerized D-glutamic acid in anthrax bacillus); helps adhere to host tissues.

Q20. [SLO 6] Which bacteria lack a cell wall?

Ans: Mycoplasma species.

Q21. [SLO 7] Differentiate between Gram positive and Gram negative cell walls.

Ans: Gram positive: thick peptidoglycan, teichoic acid. Gram negative: thin peptidoglycan, outer membrane of lipopolysaccharides, lipoproteins and phospholipids, periplasmic space.

Q22. [SLO 7] What is the periplasmic space? Why is it important?

Ans: The space between outer membrane and cytoplasmic membrane of Gram negative bacteria; beta-lactamases there break penicillin, causing antibiotic resistance.

Q23. [SLO 8] Name the three main shapes of bacteria.

Ans: Spherical (cocci), rod shaped (bacilli) and spiral (spirochetes).

Q24. [SLO 8] Differentiate between diplococci, streptococci, staphylococci and sarcina.

Ans: Diplococci: pairs; streptococci: chains; staphylococci: grape-like clumps; sarcina: packets of eight.

Q25. [SLO 8] What are vibrio and spirochetes?

Ans: Vibrio: comma shaped curved rod. Spirochetes: thin, flexible spiral bacteria, e.g. Treponema pallidum.

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Q26. [SLO 8] What is the size range of bacteria?

Ans: About 0.1 to 600 micrometre in one dimension.

Q27. [SLO 9] What is an endospore? Name two bacteria which form it.

Ans: A thick-coated resistant spore formed inside the vegetative cell; Clostridium and Bacillus.

Q28. [SLO 9] Why is endospore formation not reproduction?

Ans: There is no increase in the number of cells during the spore cycle.

Q29. [SLO 9] What does an endospore contain?

Ans: Bacterial DNA, a little cytoplasm, ribosomes, peptidoglycan, very little water and a thick keratin-like coat.

Q30. [SLO 9] Write the stages of endospore formation.

Ans: Initiation, asymmetric division (forespore), membrane development, peptidoglycan layer, spore coat, maturation (dehydration), release.

Q31. [SLO 10] How do bacteria move?

Ans: By flagella, gliding, twitching using pili, changes of buoyancy; spirochetes by twisting their helical body.

Q32. [SLO 10] Name the arrangements of flagella.

Ans: Monotrichous, amphitrichous, lophotrichous and peritrichous.

Q33. [SLO 10] What are taxes? Give examples.

Ans: Movement of bacteria toward or away from stimuli: chemotaxis, phototaxis, magnetotaxis.

Q34. [SLO 11] Describe the structure of a bacterial flagellum.

Ans: Single protein flagellin; three parts: basal body (rotatory motion), hook (joins) and hollow helical filament. About 20 nm wide, up to 20 micrometre long.

Q35. [SLO 12] Describe the genetic material of E. coli.

Ans: A single circular DNA of about 5 x 10^6 base pairs coding for about 2000 proteins; some plasmids.

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Q36. [SLO 12] What are plasmids? Which bacterium has a linear chromosome?

Ans: Accessory rings of DNA in bacteria. Borrelia (spirochete) has a linear chromosome.

Q37. [SLO 13] Classify bacteria on the basis of carbon and energy.

Ans: Autotrophs (photosynthetic, chemoautotrophic) and heterotrophs (saprotrophic, parasitic).

Q38. [SLO 13] Differentiate between photosynthetic and chemoautotrophic bacteria.

Ans: Photosynthetic use light and bacteriochlorophyll with H_2S and CO_2; chemoautotrophs oxidize inorganic substances (sulphur, ammonia) and use the energy to make carbohydrate.

Q39. [SLO 13] What are saprotrophic and parasitic bacteria?

Ans: Saprotrophs feed on dead organic matter by extracellular enzymes (Pseudomonas). Parasites take nutrients from a host and often harm it (Salmonella typhi).

Q40. [SLO 13] Differentiate between obligate and facultative parasites.

Ans: Obligate depend completely on the host (Rickettsia); facultative can live without a host (E. coli).

Q41. [SLO 13] Classify bacteria by respiration.

Ans: Aerobic (Pseudomonas), anaerobic (Spirochaeta), facultative (E. coli), microaerophilic (Campylobacter).

Q42. [SLO 14] Differentiate between photosynthesis in cyanobacteria and other photosynthetic bacteria.

Ans: Cyanobacteria: chlorophyll a, phycobilins, thylakoids, release oxygen by splitting water. Others: bacteriochlorophyll, chromatophores, no thylakoids, no oxygen, use sulphur compounds.

Q43. [SLO 15] List the phases of bacterial growth.

Ans: Lag, log (exponential), stationary and death phases.

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Q44. [SLO 15] What happens in the stationary phase?

Ans: Growth rate is zero; new cells equal dying cells due to nutrient exhaustion, toxic wastes and oxygen depletion.

Q45. [SLO 15] What is generation time?

Ans: Doubling time of bacteria: 20 minutes in E. coli to more than 24 hours in M. tuberculosis.

Q46. [SLO 16] What is binary fission?

Ans: Asexual reproduction where DNA replicates, the cell elongates and divides by a septum into two identical daughter cells.

Q47. [SLO 16] Name the methods of genetic recombination in bacteria.

Ans: Conjugation, transduction and transformation.

Q48. [SLO 16] Define conjugation. What did Lederberg and Tatum show?

Ans: Transfer of plasmid / DNA from donor to recipient through a bridge. They showed in 1946 that two mutant E. coli strains recombined to give wild type offspring.

Q49. [SLO 16] Define transduction and transformation.

Ans: Transduction: transfer of bacterial DNA by a bacteriophage. Transformation: uptake of free DNA fragments from the environment.

Q50. [SLO 17] How do bacteria recycle nitrogen?

Ans: Rhizobium fixes N_2 to ammonium; saprotrophs make ammonia; Nitrosomonas / Nitrobacter nitrify to nitrate; denitrifying bacteria release N_2.

Q51. [SLO 17] What is humus? How is it formed?

Ans: Valuable soil material formed by decomposition of dead plants and animals by bacteria and fungi; it adds nutrients and holds moisture.

Q52. [SLO 17] Write any four economic uses of bacteria.

Ans: Vinegar and lactic acid; yogurt, cheese, butter; antibiotics from actinomycetes; vitamins K and B12; biogas; single cell protein.

Q53. [SLO 17] What is biogas?

Ans: Methane (54-70%) made by bacterial decomposition of organic waste, used as fuel.

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Q54. [SLO 17] What is single cell protein?

Ans: Protein obtained from large-scale culture of bacteria, yeast, fungi or algae for food or feed.

Q55. [SLO 18] What is the contribution of Hershey and Chase?

Ans: In 1952 they used T2 bacteriophage and bacteria to prove DNA is the hereditary material.

Q56. [SLO 18] What is bioremediation?

Ans: Use of (genetically engineered) bacteria to clean pollution, e.g. oil-eating bacteria for oil spills on beaches.

Q57. [SLO 18] What are transgenic bacteria used for?

Ans: To clone genes and mass produce insulin, growth hormones and clotting factor VIII.

Q58. [SLO 19] Write the symptoms and treatment of cholera.

Ans: Profuse watery diarrhoea (Vibrio cholerae); replace fluids and electrolytes; tetracycline shortens the illness.

Q59. [SLO 19] How is typhoid prevented?

Ans: Hand washing before food handling, pasteurization of milk, proper cooking of meat, eggs and poultry, and vaccines. Causative agent: Salmonella typhi.

Q60. [SLO 19] What are DOTS and BCG?

Ans: DOTS: directly observed treatment short course for TB. BCG: vaccine of live Mycobacterium bovis inducing partial resistance to TB.

Q61. [SLO 19] Write symptoms of pneumonia and its cause.

Ans: Sudden chills, cough, pleuritic pain, rusty sputum; caused by Streptococcus pneumoniae; treated with penicillin and erythromycin.

Q62. [SLO 20] Name five bacterial plant diseases with one example each.

Ans: Leaf spot (Xanthomonas on tomato), blight (X. oryzae in rice), soft rot (Erwinia in potato), wilt (Pseudomonas solanacearum), galls (Agrobacterium tumefaciens).

Q63. [SLO 20] What are galls? Name the causes.

Ans: Localized outgrowths on plants; Rhizobium (root nodules), Pseudomonas savastanoi (olive knot), Agrobacterium tumefaciens.

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Q64. [SLO 21] What is normal flora?

Ans: Bacteria and fungi that are permanent residents of body parts such as skin, oropharynx, colon and vagina.

Q65. [SLO 22] Write any four benefits of bacterial flora to humans.

Ans: Help digestion and vitamin B and K synthesis, support the immune system, prevent colonization by pathogens, keep the vagina acidic (lactobacilli).

Q66. [SLO 23] Differentiate between disinfectants and antiseptics.

Ans: Disinfectants kill bacteria on surfaces / objects (bleach); antiseptics are for living tissue (iodine, alcohol sanitizers).

Q67. [SLO 23] What is pasteurization?

Ans: Controlled heating of milk (71 C for 15 s or 62 C for 30 min) that kills non-spore-forming bacteria without changing taste; reduces typhoid and TB.

Q68. [SLO 23] Name physical methods to control bacteria.

Ans: Sterilization, high temperature, radiation (gamma, below 300 nm), membrane filters, pasteurization, low temperature, freezing, drying.

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Chapter 6: Long Questions with Answers (SLO based)

Q1. [SLO 1, 2] Outline the taxonomic position of prokaryotes. Explain their phylogenetic position.

Ans:

- *Whittaker (1969) five kingdoms; Margulis and Schwartz (1988) kingdom Monera for prokaryotes.*
- *Molecular systematics: two domains, Bacteria and Archaea; with Eukarya forms three-domain system (Woese, 1977: archaea have separate lineage).*
- *Bacteria arose about 4 billion years ago; common ancestor of bacteria and archaea (2.5-3.2 billion years ago) probably a hyperthermophile. Domains evolved independently.*

Eubacteria	***Archaea***
Peptidoglycan present	*Absent*
Simple RNA polymerase	*Complex, like eukaryotes*
Unbranched lipid chains	*Branched lipid chains*

Q2. [SLO 3] Justify the occurrence of bacteria in the widest range of habitats.

Ans:

- *Flagella: move to good places. Pili: adhere to rocks, intestine. Endospores: survive water and nutrient shortage, heat, cold, poisons. Actinomycete filaments: bridge dry gaps in soil.*
- *Diverse nutrition and respiration (autotrophs, heterotrophs, aerobes, anaerobes) and fast binary fission.*

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Q3. [SLO 4] Draw an annotated diagram of a generalized bacterial cell.

Ans:

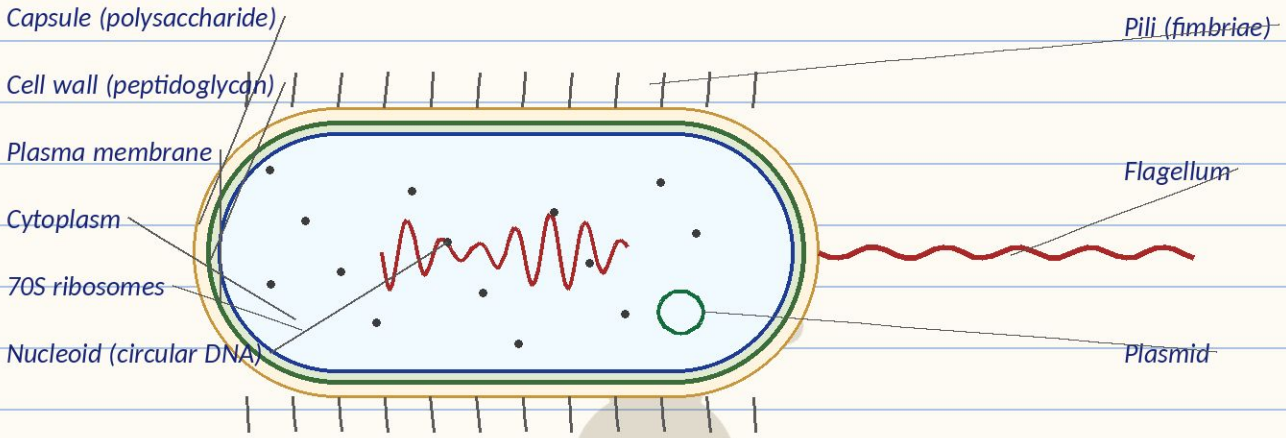


Fig: Generalized bacterial cell

- Typical cell: capsule, cell wall (peptidoglycan), plasma membrane, cytoplasm with 70S ribosomes, nucleoid with circular DNA, plasmids, and surface flagellum and pili.

Q4. [SLO 5] Justify that cyanobacteria are the most prominent photosynthetic bacteria.

Ans:

- Occur in many habitats; first land colonizers; form lichens and symbioses.
- Chlorophyll a and phycobilins (phycocyanin); thylakoid membranes; splitting of water with oxygen release; introduced oxygen into primitive atmosphere.
- One-third fix nitrogen in heterocysts (Nostoc, Anabaena) so improve soil fertility.

Q5. [SLO 6, 7] Describe the structure and chemical composition of the bacterial cell wall and other coverings. Compare Gram positive and Gram negative walls.

Ans:

- Cell wall: inner peptidoglycan (peptide + glycan) and outer membrane; capsule: polysaccharide covering (D-glutamic acid in anthrax bacillus).

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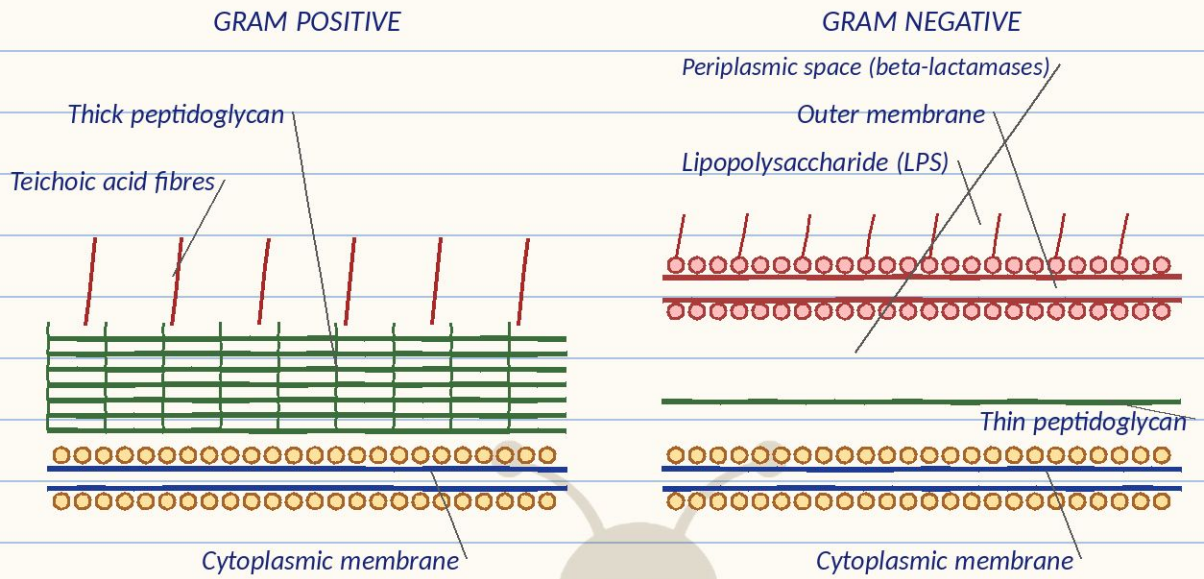


Fig: Cell wall of Gram positive and Gram negative bacteria

Gram positive	**Gram negative**
Thick peptidoglycan, teichoic acid	Thin peptidoglycan, outer membrane (LPS, lipoproteins, phospholipids)
No periplasmic space	Periplasmic space with beta-lactamases

Q6. [SLO 8] Illustrate with diagrams the diversity of shapes and sizes of bacteria.

Ans:

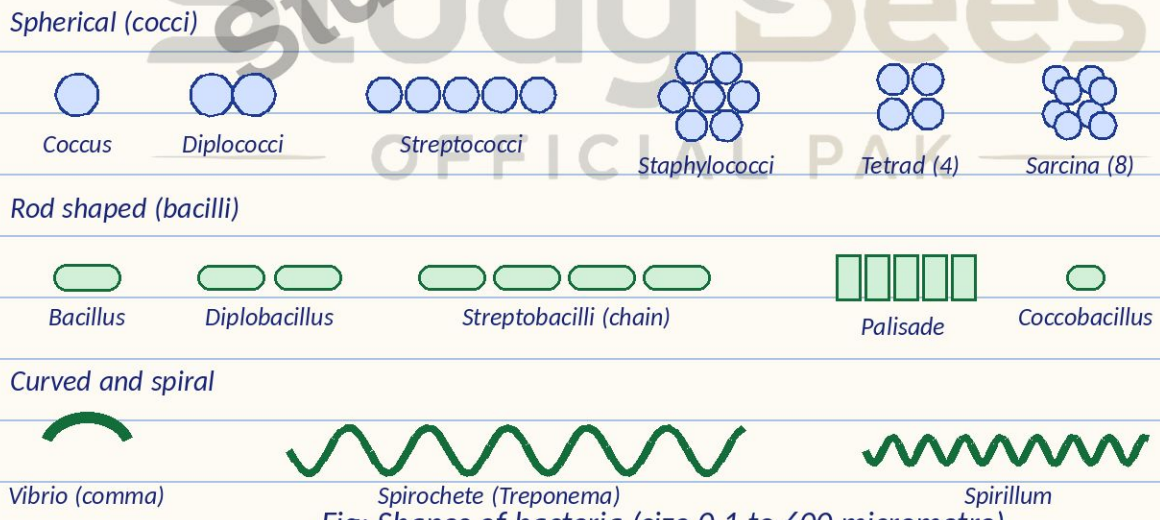


Fig: Shapes of bacteria (size 0.1 to 600 micrometre)

- *Cocci, bacilli and spirals; arrangements; size 0.1-600 micrometre.*

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Q7. [SLO 9] Justify endospore formation as a survival mechanism.

Ans:

- *Under nutrient scarcity or extreme temperature: DNA replicates; asymmetric division gives forespore; membranes, peptidoglycan cortex and spore coat form; dehydration; mother cell ruptures and spore is released.*
- *Endospore resists heat, desiccation, chemicals and poisons for long; absorbs water and germinates when conditions improve. It is survival, not reproduction.*

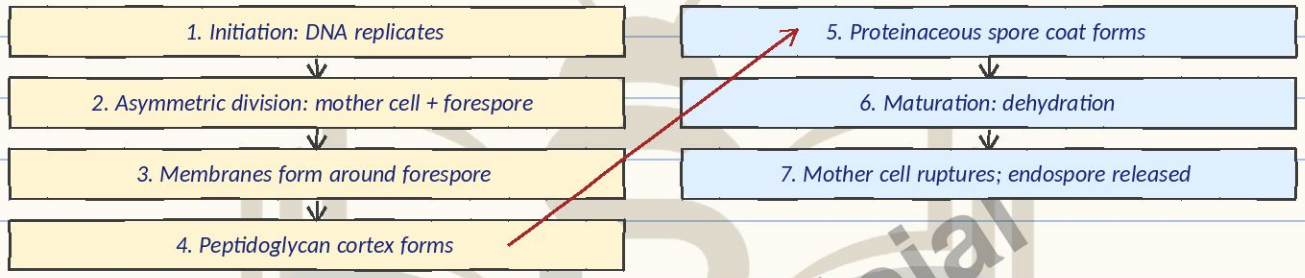


Fig: Endospore formation (not reproduction)

Q8. [SLO 10, 11] Explain motility in bacteria and describe the structure of the bacterial flagellum.

Ans:

- *Motility by flagella, gliding, twitching (pili), buoyancy changes; spirochetes twist. Flagella types: mono-, amphi-, lopho-, peritrichous; swimming and tumbling; taxes (chemo-, photo-, magneto-).*

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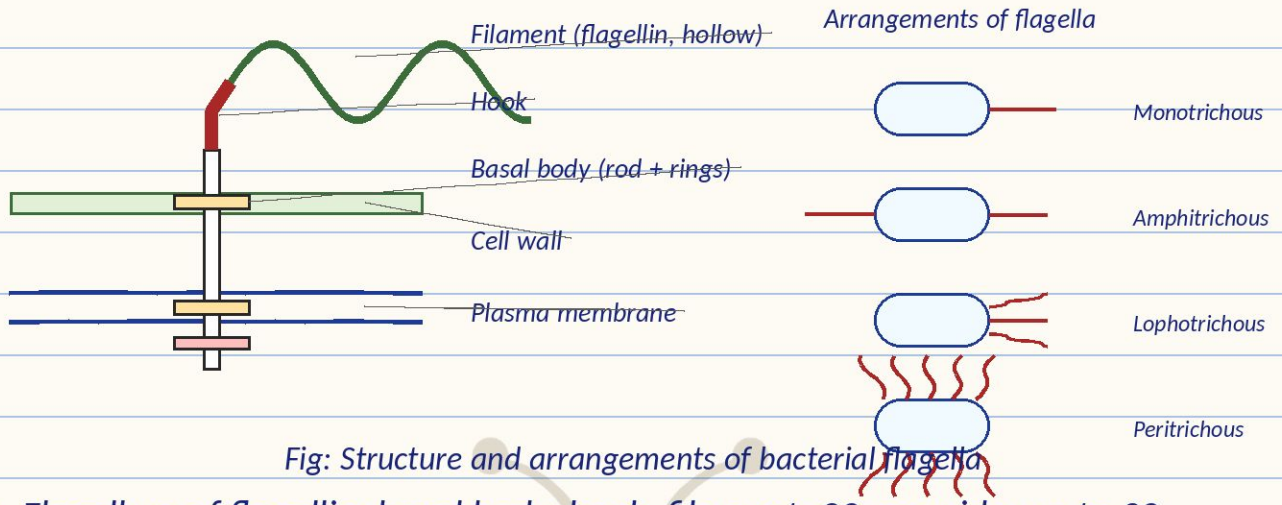


Fig: Structure and arrangements of bacterial flagella

- *Flagellum of flagellin: basal body, hook, filament; 20 nm wide, up to 20 micrometre long; rotation propels cell.*

Q9. [SLO 12] Describe the genomic organization of bacteria.

Ans:

- *Single circular chromosome; E. coli has about 5 million base pairs coding about 2000 proteins; size range 160,000 to 12.2 million base pairs; Borrelia linear.*
- *Plasmids: accessory DNA rings that can transfer between bacteria.*

Q10. [SLO 13, 14] Classify bacteria on the basis of methods of obtaining energy and carbon. Differentiate photosynthesis in cyanobacteria and other photosynthetic bacteria.

Ans:

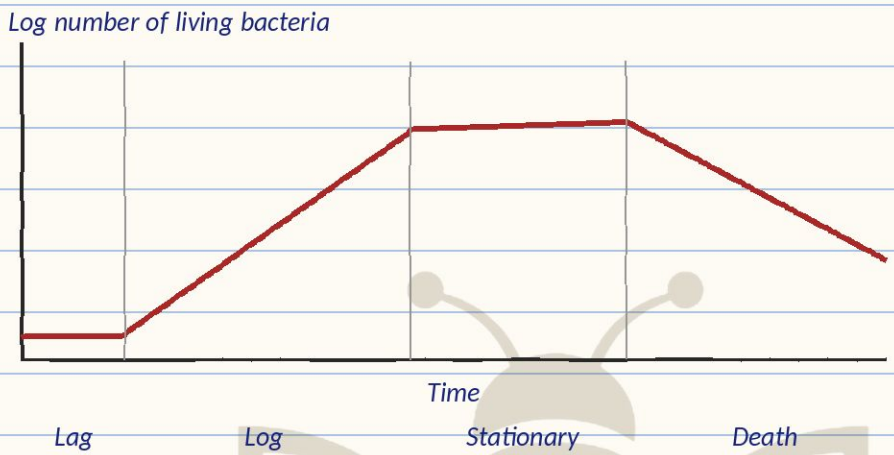
- *Autotrophs: photosynthetic (bacteriochlorophyll, H_2S) and chemoautotrophic (nitrifying, sulphur bacteria). Heterotrophs: saprotrophic (Pseudomonas, Azotobacter) and parasitic (obligate, facultative).*

Cyanobacteria	***Other photosynthetic bacteria***
Chlorophyll a, phycobilins	*Bacteriochlorophylls*
Thylakoids	*Chromatophores (membrane infoldings)*
Oxygen released (water split)	*No oxygen (H_2S etc.)*

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Q11. [SLO 15] List and explain the phases of growth of bacteria with a growth curve.

Ans:



No growth: adapting, synthesizing enzymes

Log: maximum rate

Stationary: births = deaths

Death: death rate rises

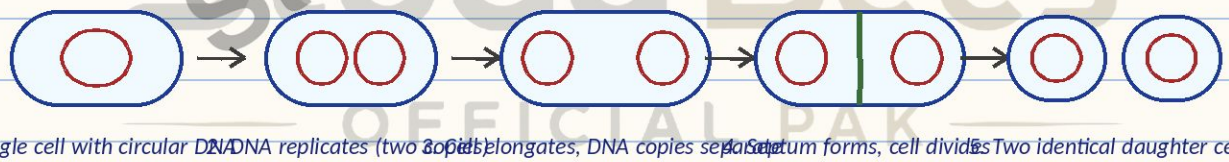
E. coli: lag 10 min, log 180 mi stationary 180-360, death 360

Fig: Growth curve of a bacterial population

- Lag: adaptation, new enzymes. Log: maximum growth. Stationary: growth rate zero (nutrient exhaustion, toxic wastes, oxygen depletion). Death: death rate increases.

Q12. [SLO 16] Describe different methods of reproduction in bacteria.

Ans:



Doubling time: 20 minutes (E. coli) to over 24 hours (M. tuberculosis)

Fig: Binary fission in bacteria

- **Binary fission:** DNA replication, cell elongation, septum, two identical daughters.
- **Genetic recombination:** conjugation (plasmid through bridge; Lederberg and Tatum 1946), transduction (via bacteriophage), transformation (uptake of free DNA).

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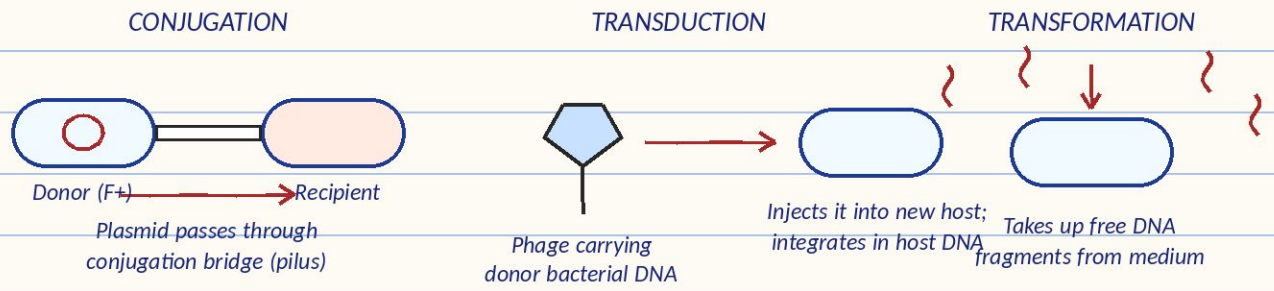


Fig: Genetic recombination in bacteria

Q13. [SLO 17, 18] Describe bacteria as recyclers of nature. Outline ecological and economic importance, and use in research and technology.

Ans:

- *Recycling: Rhizobium fixation, ammonification, nitrification (Nitrosomonas, Nitrobacter), denitrification; fermentation releases CO_2.*
- *Ecological: decomposers form humus.*
- *Economic: digestion, vitamins K and B12, industry (vinegar, acetone, lactic acid, butanol, cheese, tanning), food (yogurt, cheese, butter), antibiotics (streptomycin), biogas, single cell protein.*
- *Research and technology: Hershey-Chase experiment; insulin, growth hormone, factor VIII; frost-minus bacteria; bioremediation; phenylalanine; mineral extraction.*

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Q14. [SLO 19] Describe cholera and typhoid (symptoms, causative bacteria, treatment, prevention).

Ans:

Cholera	***Typhoid***
Vibrio cholerae	*Salmonella typhi*
Profuse watery diarrhoea	*Slow onset fever, constipation, delirium, enlarged spleen, rose spots*
Fluids and electrolytes; tetracycline	*Antibiotics for chronic carriers*

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Cholera	Typhoid
Clean water, vaccine, detect carriers	Hand washing, pasteurized milk, cooked food, vaccine

Q15. [SLO 19] Describe tuberculosis and pneumonia (symptoms, causative bacteria, treatment, prevention).

Ans:

- **TB:** Mycobacterium tuberculosis; fever 7-14 days, dry cough; later evening fever, night sweats, weight loss, bloody sputum, chest ache; multiple-drug therapy or DOTS; prompt treatment, masks, BCG vaccine.
- **Pneumonia:** Streptococcus pneumoniae; sudden chills, cough, pleuritic pain, rusty sputum; penicillin and erythromycin.

Q16. [SLO 20] Describe bacterial diseases of plants: leaf spots, blights, soft rots, wilts and galls.

Ans:

Disease	Cause and prevention
Leaf spot	Xanthomonas campestris, Pseudomonas syringae; disease-free seeds
Blight	Xanthomonas oryzae (rice), Erwinia amylovora (apple, pear); remove diseased plants
Soft rot	Erwinia atroseptica (potato); remove diseased plants
Wilt	Pseudomonas solanacearum; right planting date and spacing
Galls	Agrobacterium tumefaciens, Rhizobium, P. savastanoi; crop rotation

Q17. [SLO 21, 22] Define normal flora and describe its benefits to humans.

Ans:

- Normal flora: permanent bacteria and fungi of skin, oropharynx, colon and vagina (e.g. E. coli, Lactobacillus, Clostridium, Enterococcus faecalis).
- Benefits: digestion and vitamins B and K; immune support; competitive

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exclusion of pathogens; short chain fatty acids; gut-brain axis; acidic vagina; detoxification; metabolism; immune development; recovery after antibiotics; synthesis of biotin and folate.

Q18. [SLO 23] List the chemical and physical methods used to control harmful bacteria.

Ans:

- ***Chemical:*** *disinfectants (bleach), antiseptics (iodine, alcohol), preservatives (sodium benzoate), chlorine in water treatment, chemotherapy (sulfonamides).*
- ***Physical:*** *sterilization, high temperature (moist / dry heat), radiation (gamma), membrane filter, pasteurization (71 C for 15 s), low temperature, freezing (-10 to -18 C), drying.*

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